Aaron Kleiman

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Education

Queen's University Kingston, ON
Bachelor of Applied Science in Computer Engineering *Sept 2023 – Apr 2028*

- Relevant Coursework: Operating Systems, Computer Architecture, Microprocessor Systems, Data Structures & Algorithms, Databases, Data Science, Digital Systems Engineering, OOP, Calculus, Linear Algebra, Discrete Math, Probability

Experience

Systems Software Engineer Intern (Datacenter GPU Validation) May 2026 – Present
AMD *Markham, ON*

- Validate and debug pre-release **firmware, drivers, and ROCm software** on bare-metal **MI300X, MI300A, and MI450 GPU clusters**, identifying failures across **AI/ML, graphics, stability, and error-recovery** workloads before release
- Develop domain-specific triage agents within a **multi-agent framework** that orchestrates end-to-end GPU regression and hardening cycles, automating failure triage and debugging across validation domains
- Develop **Python test automation** and **YAML configurations** for containerized **HPC and AI workloads**, integrating BabelStream, HPCG, LAMMPS, and Megatron-LM into **Jenkins** CI/CD for repeatable regression coverage
- Built a production **Python triage pipeline** triggered automatically at job completion, reducing per-job failure analysis from **40 minutes** of manual **dmesg and system-log review** to a **10-minute structured report**
- Engineered an **AI agent** for reliability, availability, and serviceability (RAS) failures that automates triage and root-cause classification, generating structured reports for comparison across regression runs

Software Engineer Intern May 2025 – Aug 2025
Tallysight *San Diego, CA (Remote)*

- Built a **Python ETL pipeline** automating ingestion of **100+ analyst articles**, replacing an **8-hour** manual task
- Automated editorial data entry using **LLM-powered n8n and Slack workflows**, reducing hours of work to **minutes**
- Developed **Retool AI agents** to update player and team assets, replacing manual search and **MongoDB** uploads

Software UI Lead Jul 2025 – May 2026
Queen's Space Engineering Team *Kingston, ON*

- Led the design of a **Next.js** rover dashboard integrating **ROS 2 nodes over WebSockets** for real-time telemetry
- Led a software sub-team and standardized **GitLab** code reviews and merge checklists, improving quality and onboarding

Undergraduate Teaching Assistant Sept 2025 – Dec 2025
Queen's University *Kingston, ON*

- Coordinated weekly **C programming** labs for **200+** first-year students, providing debugging support, feedback, and grading

Software Developer (Volunteer) Oct 2024 – May 2025
Swarmed (beeswarmed.org) *Mountain View, CA (Remote)*

- Redesigned **UI/UX** for a beekeeper–public platform, contributing to the protection of **250M+ honey bees**
- Built dashboard survey tooling with **JavaScript** and **Tally Forms**, streamlining user-feedback collection
- Automated user interactions and backend operations through **Bubble.io workflows** and **REST API integrations**

Projects

OdysseyWalk | *Next.js 14, TypeScript, Google Maps API, LLM APIs, WebSockets*

- Won **QHacks 2026** with a voice-first tour app generating route-aware itineraries through LLM and Google Maps APIs
- Built **WebSocket** TTS/STT streaming and GPS-triggered narration for real-time, hands-free guided walks

Autonomous Taxi Robot | *Python, Raspberry Pi 4, Coral Edge TPU, asyncio, TensorFlow Lite*

- Designed and built an autonomous mini-taxi with an **event-driven Python stack**, FSM, and **20 Hz asyncio** loop
- Implemented **Coral Edge TPU** perception, PID control, and **heading-aware A* path planning** on a 37-node road graph

32-Bit RISC Processor | *Verilog, ModelSim, Finite State Machines, Computer Architecture*

- Designed a **32-bit single-bus RISC CPU in Verilog** with 16 registers, an ALU, memory, and FSM-based control
- Verified arithmetic, memory, branch, jump, I/O, multiply, and divide instructions with **ModelSim testbenches**

Technical Skills

Languages: Python, C, C++, Bash/Shell, JavaScript/TypeScript, Java, SQL, VHDL, Assembly
Systems & Infrastructure: Linux/Unix, Docker, Git, GitLab, Jenkins, CI/CD, YAML, AWS, GCP
GPU & Accelerators: CUDA, ROCm, HPC, regression testing, test automation, TensorFlow Lite, Coral Edge TPU
AI/ML: AI agents, multi-agent orchestration, NumPy, Pandas, OpenAI API, Claude Code, Cursor
Web & Data: React, Next.js, Node.js, Express, HTML/CSS, WebSockets, REST APIs, MySQL, MongoDB, n8n, Bubble.io
Hardware & Embedded: Arduino, Raspberry Pi, ROS 2, FPGA, JTAG, LTspice, Altium, Oscilloscope